

**UCSC****University of Colombo, Sri Lanka***University of Colombo School of Computing***DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**Academic Year 2023 — 3rd Year Examination — Semester 6**IT6505 (R) — Middleware Architecture
(Repeat Paper)***Structured Question Paper*
(2 Hours)**To be completed by the candidate****Index Number**

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Important Instructions

- The duration of the paper is **2 hours**.
- The medium of instructions and questions is English. Students should answer in the medium of English language only.
- This paper has **4 questions** on **12 pages**. Answer **all** questions.
- All questions carry **equal** marks.
- Write your answers **only on the space provided** on this question paper.
- Do not tear off any part of this question paper. Under no circumstances may this paper (or any part of this paper), used or unused, be removed from the Examination Hall by a candidate.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are **not allowed**.
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**To be completed by
the examiners**

1	
2	
3	
4	
Total	

1)

- (a) Briefly describe one (01) advantage and one (01) disadvantage of a distributed system compared to a centralized system.

(6 marks)**ANSWER IN THIS BOX**

Disadvantages:

- Distributed systems allow the sharing of information and resources, and may be composed of small, cost-effective computers that combine their processing power.
- Large computational problems can be solved at low cost by temporarily using the combined power and memory of many computers.

Disadvantage:

Distributed systems are harder to manage because we have to address issues such as independent failures, unreliable communication, and insecure communication.

➔ 1 mark for each definition + 2 mark for explanation

- (b) State three (03) benefits of using a layered network model for the design and implementation of networked systems.

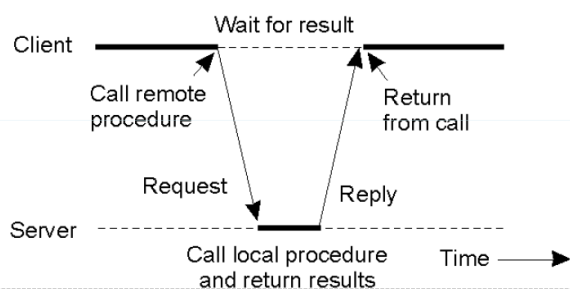
(3 marks)**ANSWER IN THIS BOX**

- It facilitates troubleshooting
- It breaks the complex process of networking into more manageable chunks
- It allows layers developed by different vendors to interoperate.

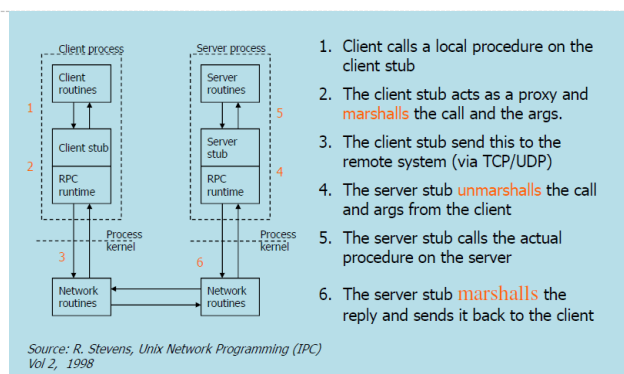
This image shows a single sheet of white paper with horizontal blue or grey ruling lines, typical of notebook paper. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

- (c) Using a diagram, describe the operation of the Remote Procedure Call (RPC) mechanism between a Client and a Server Program.

(6 marks)

ANSWER IN THIS BOX

OR



- (d) What is meant by the term ‘reflective middleware’? Explain briefly.

(5 marks)

ANSWER IN THIS BOX

- A configurable and reconfigurable middleware platform that uses reflection as a principled mechanism to dynamically adapt middleware behaviour to changing environmental context.
- In computer science, reflection is the ability of a computer program to examine, introspect, and modify its own structure and behavior at runtime.
- The middleware contains an interface that supports operations to discover the internal operation and structure of the middleware platform (e.g. protocols and management structures being deployed) and to make changes at run-time.

- (e) What is meant by *Data Access Middleware*? Give an example for a data access middleware API. (5 marks)

ANSWER IN THIS BOX

Provide a uniform interface to relational and nonrelational data using sql. Requests to access data from a dbms are sent to a data access driver rather than directly to the dbms. The data access driver converts the sql statement into the sql supported by the dbms and then routes the request to the dbms.

→ 4 mark

The two leading data access middleware are the open database connectivity (odbc) supported by microsoft and the java database connectivity (jdbc) supported by oracle.

→ 1 mark

2)

(a) List *four (04)* functions expected of middleware services.

(4 marks)

ANSWER IN THIS BOX

- Locate transparently across the network, thus providing interaction with another service or application
- Filter data to make them friendly usable or public via anonymization process for privacy protection
- Be independent from network services
- Add complementary attributes like semantics
- Be reliable and always available

(b) Explain the difference between a *vertical connection* and a *horizontal connection* in a layered middleware architecture.

(6 marks)

ANSWER IN THIS BOX

- Vertical connection refers to the connection of the different layers in a multitiered architectures across multiple machines.
- In principle, each layer is implemented on a different machine.
- Horizontal connection deals with the connection of a single layer across multiple machines, such as distributing a single database.

- (c) Briefly explain the term *Message Broker Pattern* in message oriented middleware.

(6 marks)

ANSWER IN THIS BOX

Message brokers are a building block of Message oriented Middleware. Pattern is a general repeatable solution to a commonly occurring problem in software design. A message broker pattern is an architectural pattern for message validation, transformation and routing.

- It mediates communication amongst applications, minimizing the mutual awareness that applications should have of each other in order to be able to exchange messages, effectively implementing decoupling.
- The purpose of a broker is to take incoming messages from applications and perform some action on them..

- (d) Explain the functionality of the code below.

(9 marks)

```
package example.hello;

import java.rmi.registry.Registry;
import java.rmi.registry.LocateRegistry;
import java.rmi.RemoteException;
import java.rmi.server.UnicastRemoteObject;
```

```

public class Server implements Hello {

    public Server() {}
    public String sayHello() {
        return "Hello, world!";
    }

    public static void main(String args[]) {

        try {
            Server obj = new Server();
            Hello stub = (Hello) UnicastRemoteObject.exportObject(obj, 0);
            Registry registry = LocateRegistry.getRegistry();
            registry.bind("Hello", stub);
            System.err.println("Server ready");
        } catch (Exception e) {
            System.err.println("Server exception: " + e.toString());
            e.printStackTrace();
        }
    }
}

```

ANSWER IN THIS BOX

This code is brought from a Java RMI implementation. A "server" class, in this context, is the class which has a main method that creates an instance of the remote object implementation, exports the remote object, and then binds that instance to a name in a Java RMI registry. The class that contains this main method could be the implementation class itself, or another class entirely. server obtains a stub for a registry on the local host and default registry port and then uses the registry stub to bind the name "Hello" to the remote object's stub in that registry. Once a remote object is registered on the server, callers can look up the object by name, obtain a remote object reference, and then invoke remote methods on the object.

The static method `LocateRegistry.getRegistry` that takes no arguments returns a stub that implements the remote interface `java.rmi.registry.Registry` and sends invocations to the registry on server's local host on the default registry port of 1099. The bind method is then invoked on the registry stub in order to bind the remote object's stub to the name "Hello" in the registry.

3.

- (a) Briefly discuss three (03) practical benefits of using in-process servers (DLLs) in DCOM.

(6 marks)

ANSWER IN THIS BOX

In-process servers (Dynamic Link Libraries or DLLs) can be dynamically load into the program when they are required.

Practical Usages:

- No additional requirement of memory at runtime or disk storage as a common binary file will be there,
- All the application or clients will be using one single version of the common code provided by the DLL,
- Operating system loads only one instance of the DLL when the first application/client loads it then for every subsequent application it shares the memory pages of the DLL with their process address space. Thus, there is no unnecessary memory overload in the system.

- (b) What is meant by *Dynamic Invocation* in CORBA based development?

(5 marks)

ANSWER IN THIS BOX

When using dynamic invocation, the CORBA client application can dynamically build operation requests for a CORBA object interface that has been stored in the Interface Repository. This supports deferred synchronous communication, where the CORBA client application sends the request and it is free to do other work. Dynamic Invocation also enables a CORBA client

application to invoke a method on a CORBA object whose type was unknown at the time the CORBA client application was written.

- (c) Briefly describe three (03) differences between CORBA and RMI from a developer's perspective.

(6 marks)

ANSWER IN THIS BOX

- RMI is a Java-specific technology and runs on top of JVM. CORBA has implementations for many languages as well as CORBA services can be executed on many different platforms.
- RMI uses Java interfaces. CORBA uses IDL (Interface Definition Language) to separate interface from implementation. Therefore, CORBA developers can create different implementations based on the same interface.
- CORBA objects are not garbage collected. Once a CORBA object is created, it continues to exist until programmer should decide when to get rid of it. RMI objects are garbage collected automatically.
- In CORBA, the data types do not always map exactly to the types used by the programming language (e.g. a long in IDL is an int in Java).
- RMI programs can download new classes from remote JVMs. CORBA doesn't have a code sharing mechanism.
- RMI can be operated over IIOP (the protocol used by CORBA). Through RMI-IIOP it is possible to achieve interoperability.

- CORBA supports in and out parameters, while RMI does not since local objects are passed by copy and remote objects are passed by reference.

- (d) Define and then briefly describe three (3) functionalities provided by a *Portable Object Adapter (POA)* in CORBA.

(4 marks)

ANSWER IN THIS BOX

In CORBA, object adapter connects a request using an object reference with the proper code to service that request. The Portable Object Adapter (POA) is a particular type of object adapter that provides following functionalities:

- ☐ POA allows programmers to construct object implementations that are portable between different ORB products.
- ☐ It supports for objects with persistent identities.
- ☐ It supports transparent activation of objects.
- ☐ Associate policy information with objects.
- ☐ Allow multiple distinct instances of the POA to exist in one ORB.

- (e) Describe what is meant by the approach *Hypermedia as The Engine of Application State (HATEOAS)*.

(4 marks)

ANSWER IN THIS BOX

The principle is that a client interacts with a network application entirely through hypermedia provided dynamically by application servers. A REST client needs no prior knowledge about how to interact with any particular application or server beyond a generic understanding of hypermedia.

4.

- (a) Refer the given code below and answer the Questions (i) and (ii).

```
<?xml version="1.0"?>
<soap:Envelope
xmlns:soap="http://www.w3.org/2003/05/soap-envelope/"
soap:encodingStyle="http://www.w3.org/2003/05/soap-encoding">
<soap:Body>
  <m:GetName xmlns:m="https://www.abcschoolsl.com/names">
    <m:StudentName>22001722</m:StudentName >
  </m: GetName >
</soap:Body>
</soap:Envelope>
```

- i. Briefly explain the meaning of the **SOAP message** given above.

(3 Marks)

- ii. Write down the expected **SOAP response message** for the message in (i).

(5 Marks)

ANSWER IN THIS BOX

(i)

The above SOAP message requests the Name of a person. m: GetName and the m:StudentName elements above are application-specific elements. They are not a part of the SOAP namespace.

→ 3 mark

(ii) A SOAP response could look something like this:

```
<?xml version="1.0"?>

<?xml version="1.0"?>
<soap:Envelope
xmlns:soap="http://www.w3.org/2003/05/soap-envelope/"
soap:encodingStyle="http://www.w3.org/2003/05/soap-encoding">
```

```
<soap:Body>
  <m:GetNameResponse xmlns:m="https://www.abcschoolsl.com/names">
    <m:StudentName>Jhon</m:StudentName >
  </m: GetNameResponse >
</soap:Body>
</soap:Envelope>
```

→ 6 mark

- (b) Identify the most suitable **HTTP verb** to be used in each of the following RESTful URIs.

(5 marks)

ANSWER IN THIS BOX

URI	HTTP Method
/book/{id}	<u>GET</u>
/book/add	<u>POST</u>
/book/delete/{id}	<u>DELETE</u>
/book/getAll	<u>GET</u>
/book/update/{id}	<u>PUT</u>

- (c) The following piece of code was taken from a **Controller class** in a Restful backend application.

Explain The *functionality* of the code given below.

(8 marks)

```
@RestController
class RouteController {
    private final RouteRepository route;
```

```

RouteController(RouteRepository route) {
    this.route = route;
}
@PutMapping("/route/addWaypoint/{id}")
Waypoint replaceWaypoint(@RequestBody Waypoint newWaypoint,
@PathVariable Long id) {
    return route.findWaypointById(id)
        .map(waypoint -> {
            waypoint.setName(newWaypoint.getName());
            waypoint.setCoordinates(newWaypoint.getCoordinates());
            return trail.saveWaypoint(waypoint);
        })
        .orElseGet(() -> {
            newWaypoint.setId(id);
            return trail.saveWaypoint(newWaypoint);
        });
}
>DeleteMapping("/route/delete/{id}")
void deleteRoute(@PathVariable Long id) {
    route.deleteById(id);
}
}

```

ANSWER IN THIS BOX

@RestController indicates that the data returned by each method will be written straight into the response body instead of rendering a template.

→ 2 mark

An RouteRepository is injected by constructor into the controller.

→ 2 mark

There are two operations (@PutMapping and @DeleteMapping, corresponding to adding a new waypoint to a route, and DELETE trails).

→ 2 mark

During the update if the waypoint Id is not existing, a new waypoint will be created in the database.

→ 2 mark

- (d) The following service call was found in a piece of code in a Controller class of a Restful backend application.

```
Route route = routeService.findRouteById(routeId);
```

Briefly explain the expected functionality of the above *service call*.

(3 marks)

ANSWER IN THIS BOX

Service classes implements the business logic (methods to create, retrieve, and manipulate data) for a particular entity. Here the route service.findRouteById method will output a route object after searching for a route using a routeID.
